

Pinching Visuo-Haptic Tactile Display: A Dynamic Cloth Texturization Via Visually Enhanced Tactile Sensation

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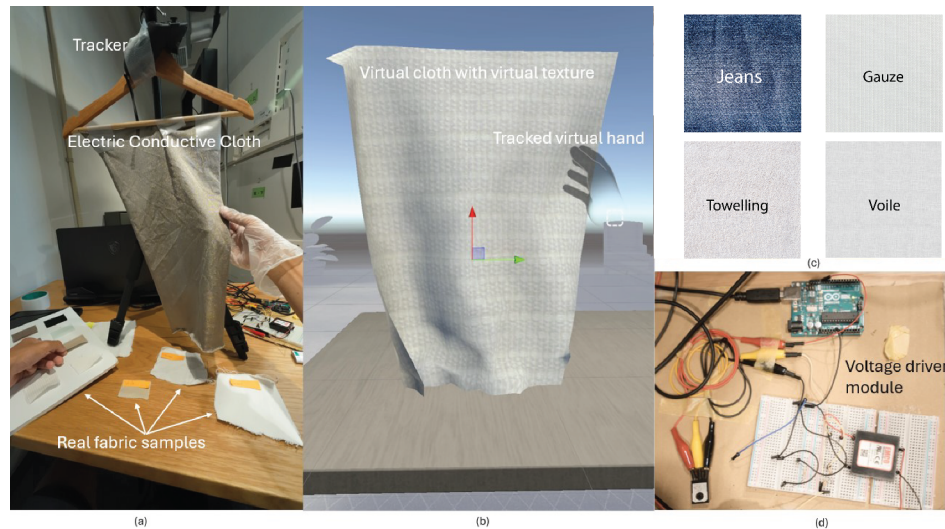


Figure 1: (a) shows our proposed pinching visuo-haptic display system. The electric conductive cloth(ECC) is hung on a hanger with a VR tracker attached to it on the top. The user is currently wearing a VR HMD while touching the ECC with a glove on the right hand for safety reasons and is comparing the tactile sensation to the real fabric on the right. (b) shows the same setup in the VR modal. In VR, a virtual cloth is placed at the position of the ECC in real world. We assigned the texture of the virtual cloth with our target fabric texture. (c) shows the fabric textures of Jeans, Gauze, Towelling, and Voile that are used to render on the virtual cloth. (d) shows the voltage driver module used to apply various voltage conditions.

1 INTRODUCTION

The adoption of Virtual Reality (VR) in e-Commerce, particularly in the fashion sector, has seen unprecedented growth [2]. One of the biggest appeals of VR is its immersiveness, which could range from teleporting users to environments with engaging shopping experiences [9] to allowing virtual try-ons [7].

One of the most important factors influencing purchasing decision making in fashion is the fabric [6] of the garment. Unlike design, which is a visual first property, fabric is often evaluated by touching the garment directly before buying. [1] suggested that having multisensory capabilities in addition to VR could greatly increase the shopping experience.

A system called "Telexiles", which is an interface that can remotely transmit tactile sensations of textiles, is proposed by [4]. It is a system that can scan and learn the profiles of different fabrics and present them to a user remotely via a rotating mechanism using a predefined set of real fabrics. Although it gives an accurate tactile sensation, it is limited by the physical space constraint. Another tactile display that generates tactile sensation, the "Pinching Tactile Display" [3], uses an electric conductive cloth(ECC) to emulate

the tactile sensation of a set of fabrics by altering the voltages and frequencies applied to the ECC. They reported that users could perceive a difference in the thickness, stiffness, and roughness of the fabrics to a certain degree but not in the temperature of the fabrics. The authors explored the potential for fabric emulation; however, the resolution of the PTD was restricted, limiting their investigation to only determining whether the tactile sensation of the ECC could be perceived as similar to that of cloth.

The idea of multisensory VR coupled with haptics is, in fact, one of the research topics that has advanced rapidly in recent years. To achieve a realistic tactile feedback for the users, we aim to enhance the tactile feedback through the addition of visual cues. We propose a method to overlay the target fabric visuals in the VR onto the ECC in the real world, while the ECC's tactile feedback is tuned according to the visual. In this paper, we hypothesize that by overlaying a visually correct target fabric on the ECC, we could enhance the tactile sensation of that particular fabric to feel like its real-world counterpart. Our main contributions include the following.

- Proposing a visually enhanced dynamic cloth texturization system using VR and ECC.
- Conducting a user study with the hypothesis that the tactile sensation of ECC could be enhanced towards the target fabric with immersive visuals of VR.

2 METHOD AND IMPLEMENTATION

For the VR modality, we implement the pinching visuo-haptic display in Unity. We used Meta Quest Pro as our VR HMD of choice.

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Term	Roughness	Stiffness	Thickness	Warmth
volt	$4.00 \times 10^{-6}*$	$2.92 \times 10^{-3}*$	$8.00 \times 10^{-6}*$	0.0864
VR	0.0669	0.525	0.316	0.515
volt $\times$ VR	0.585	0.483	0.915	0.990

Table 1: ANOVA results for roughness, stiffness, thickness, and warmth evaluations (voltage and VR interactions, p-value). * indicates significant results ($p < 0.05$). This indicates a marginally significant difference in roughness perception with VR ($Pr > F = 0.0669$), suggesting that VR may affect roughness perception

The visual of the virtual cloth, such as the textures and movements, is important to seamlessly blend with the real-world object. We used high-resolution fabric textures as shown in Figure1(c) from Quixel Megascan library to visualize our target fabric to achieve a realistic look.

Cloth Simulation We implemented the cloth simulation solver based on the spring-mass system [5] with a custom particle collider system to allow interaction with the hand tracking of the Meta Quest Pro. We manually tuned the spring stiffness and damping coefficient for each of the fabrics used in the user experiment to match their real-world counterparts: 1)denim, 2)gauze, 3)towel, and 4)voile.

World blending In order to align the object in the virtual world with the real world, we used the Meta Quest Pro controller as the tracker. Then we sync the position of the simulated cloth to the tracker object in Unity as we do in the real world: we hang the ECC on a hanger taped with a controller on to a tripod. The virtual space and table are also scaled according to their respective real-world counterparts so that the user can interact with the cloth seamlessly as if there exists a physical world.

3 USER STUDY DESIGN

We conducted a series of user studies to validate the hypothesis mentioned in the previous section. In this section, we will describe our user study design that incorporates visuo-haptic feedback of cloths and fabrics. Let us revisit our hypotheses:

Hypothesis Participants can experience changes in the roughness, stiffness, thickness, and warmth of the ECC that align with the visuals of the virtual fabric types compared to those without visuals.

The user study consists of two sessions for each participant, one wearing a VR HMD and one without wearing a VR HMD as a control. During each session, each participant is asked to touch four types of fabrics: 1) denim, 2) gauze, 3) towel, and 4) voile. For each of those types of fabric, the types of tactile sensations that we want to investigate are roughness, stiffness, thickness, and warmth. Participants are required to stroke the ECC with a voltage range applied in random order: [50V, 100V, 150V, 200V, 250V, 300V] for each type of fabric, for each modal. After experiencing each of the voltage conditions, participants are asked to answer a questionnaire on an Ipad placed on the same table. The questionnaire consists of four 1- to 5-likert scales, each corresponding to rough (1) to smooth (5), stiff(1) to flexible(5), thick(1) to thin(5) and warm(1) to cool(5).

4 EVALUATION

To analyze how the introduction of visual cues in VR alters tactile perception, we conducted a statistical analysis of participants' Likert scale responses. The experimental design consisted of participants touching a cloth sample in two conditions: with and without VR HMD. Because our data followed a design within participants and normality could not be assumed, we applied an aligned rank transform (ART) and performed a three-way analysis of variance to compare VR and non-VR conditions for each tactile property [8].

Our analysis (Table 1) revealed a marginally significant difference in roughness perception with VR ($Pr > F = 0.0669$), suggesting that VR may affect roughness perception and warrants further investigation. These results suggest that VR integration may alter the

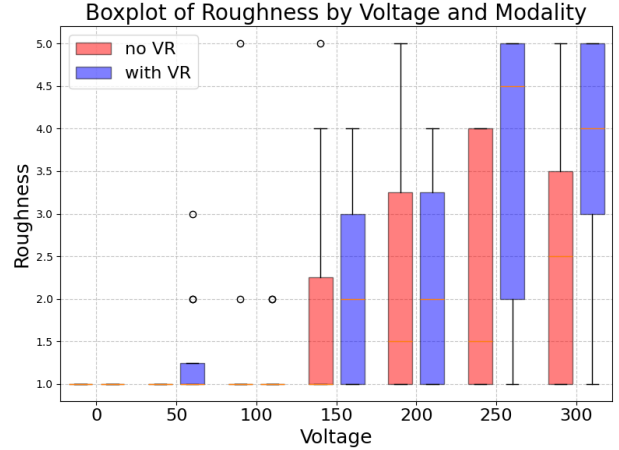


Figure 2: This graph shows the perception of roughness when participants touched the cloth while looking at a towel placed in a virtual reality (VR) room and when they touched the cloth without looking at the VR room. The results show that participants tended to perceive the roughness more when they touched the towel while looking at it than when they did not look at the VR room. This suggests that the tactile sensation of roughness may be enhanced when looking at a rough-textured cloth.

perception of certain cloth properties, particularly roughness, and may warrant larger studies and more refined experimental designs.

For texture-specific effects, we did the same analysis for each texture. The results showed that for towel (p = 0.00547), VR had a significant effect on the perception of roughness, while no significant differences were observed for denim (p = 0.147), gauze (p = 0.0671) and voile (p = 0.645). This suggests that texture-rich visuals, such as towel, may affect the roughness of cloth.

To further investigate this, we analyzed changes in roughness perception at different voltage levels for towel. As shown in Figure 2, participants perceived a greater sense of roughness when terry cloth was displayed. This finding suggests that displaying the texture of the terry cloth can enhance the perception of roughness.

These results suggest that by visually emphasizing texture, it may be possible to convey similar roughness sensations at lower voltages. A limitation of the Pinching Tactile Display is its reliance on high voltage to produce strong tactile sensations, which poses a risk of electrical shock during use. The current results suggest that this limitation can be overcome by integrating the visual appearance of rough textures into the Pinching Tactile Display, allowing the transmission of roughness sensations at a lower voltage range.

5 CONCLUSION

In this study, we propose a visually enhanced dynamic cloth texturing system using virtual reality (VR) to improve haptic perception of cloth in virtual environments. We found that displaying fabrics with rougher appearance facilitated the perception of roughness. This suggests that pinching tactile displays can convey roughness sensations at lower voltages and may overcome some of the limitations of current pinching tactile display technology.

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REFERENCES

- [1] L. Berg and J. Vance. Industry use of virtual reality in product design and manufacturing: a survey. *Virtual Reality*, 21, 03 2017. doi: 10.1007/s10055-016-0293-9
- [2] D. Grewal, S. Motyka, and M. Levy. The evolution and future of retailing and retailing education. *Journal of Marketing Education*, 40:027347531875583, 02 2018. doi: 10.1177/0273475318755838
- [3] T. Kitagishi, H. Hiraki, H. Nakamura, Y. Ishiguro, and J. Rekimoto. Pinching tactile display: A cloth that changes tactile sensation by electrostatic adsorption. In *Proceedings of the 2024 International Conference on Advanced Visual Interfaces, AVI '24*. Association for Computing Machinery, New York, NY, USA, 2024. doi: 10.1145/3656650.3656690
- [4] T. Kitagishi, Y. Hiroi, Y. Watanabe, Y. Itoh, and J. Rekimoto. Telexiles: End-to-end remote transmission of fabric tactile sensation. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology, UIST '23*. Association for Computing Machinery, New York, NY, USA, 2023. doi: 10.1145/3586183.3606764
- [5] T. Liu, A. W. Bargteil, J. F. O'Brien, and L. Kavan. Fast simulation of mass-spring systems. *ACM Trans. Graph.*, 32(6), nov 2013. doi: 10.1145/2508363.2508406
- [6] M. Nasution, Y. Rossanty, K. Ariffin, and N. Zaini. An empirical examination of the factors influencing consumer's purchase intention toward online shopping. *Journal of Business Retail Management Research*, 13, 07 2019. doi: 10.24052/JBRMR/V13IS04/ART-02
- [7] M. Ricci. Exploiting virtual reality for enhancing the shopping experience in the fashion industry: Between interaction and perception. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 938–941, 2022. doi: 10.1109/ISMAR-Adjunct57072.2022.00210
- [8] J. O. Wobbrock, L. Findlater, D. Gergle, and J. J. Higgins. The aligned rank transform for nonparametric factorial analyses using only anova procedures. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, CHI '11*, p. 143–146. Association for Computing Machinery, New York, NY, USA, 2011. doi: 10.1145/1978942.1978963
- [9] H. Wu, Y. Wang, J. Qiu, J. Liu, and X. Zhang. User-defined gesture interaction for immersive vr shopping applications. *Behaviour Information Technology*, 38:1–16, 12 2018. doi: 10.1080/0144929X.2018.1552313